EXPT NO: 9	A PYTHON PROGRAM TO IMPLEMENT THE UNCONSTRAINED OPTIMIZATION- NONLINEAR LEAST SQUARES
DATE:	

AIM:

The objective of this lab is to understand and implement optimization techniques for solving nonlinear least squares problems without constraints.

PROCEDURE:

1.PROBLEM FORMULATION:

- Define the nonlinear least squares objective function f(x).
- Specify the data points and the model that relates to the data.

2.CHOOSE AN OPTIMIZATION LIBRARY:

- Select an appropriate optimization library (e.g., scipy.optimize in Python).

3.IMPLEMENTING THE OPTIMIZATION:

- Set up the objective function to be minimized.
- Define any constraints (if the problem extends to constrained NLS).

4.RUNNING OPTIMIZATION:

- Execute the chosen optimization algorithm.
- Monitor convergence and iterate as necessary.

5.POST-OPTIMIZATION ANALYSIS:

- Evaluate and interpret the results obtained from the optimization.
- Compare with initial assumptions and data.

PROBLEM -1: TO ESTIMATE THE PARAMETERS OF AN EXPONENTIAL DECAY MODEL USING NONLINEAR LEAST SQUARES.

Parameter	Value
`t`	Linearly spaced values from 0 to 1 (50 points)
`y_true`	[2.0, 0.5]
`x0` (Initial Guess)	[1.0, 1.0]
`Noise`	0.1 * `np.random.randn(50)`
`Model`	`X[0] * np.exp(-X[1] * t)`

CODE

```
import numpy as np from scipy.optimize import least_squares

def model(X, t): return X[0] * np.exp(-X[1] * t)

def residual(X, t, y): return model(X, t) - y

np.random.seed(123)

t = np.linspace(0, 1, 50)

y_true = [2.0, 0.5] y = model(y_true, t) + 0.1 * np.random.randn(50)

x0 = [1.0, 1.0]

result = least_squares(residual, x0, args=(t, y))

print("Optimized parameters:", result.x)
```

OUTPUT

optimized parameters: [1.98254019 0.47917343]

PROGRAM 2

Estimate the parameters of the exponential model:

$$y=a\cdot e-bty=a\cdot e-bt$$

Given:

- True parameters: $a=3.0,\ b=1.2a=3.0, b=1.2$
- Add noise to the data

```
import numpy as np from scipy.optimize import least_squares

def model(X, t): return X[0] * np.exp(-X[1] * t)

def residual(X, t, y): return model(X, t) - y

np.random.seed(10)

t = np.linspace(0, 2, 50)

y_true = [3.0, 1.2] y = model(y_true, t) + 0.1 * np.random.randn(50)

x0 = [1.0, 0.5]

result = least_squares(residual, x0, args=(t, y))

print("Optimized parameters:", result.x)
```

OUTPUT

Optimized parameters: [3.05 1.18]

RESULT

Thus, solving any Unconstrained Optimization using the Non-Linear Least Squares Algorithm in python has been implemented and executed successfully